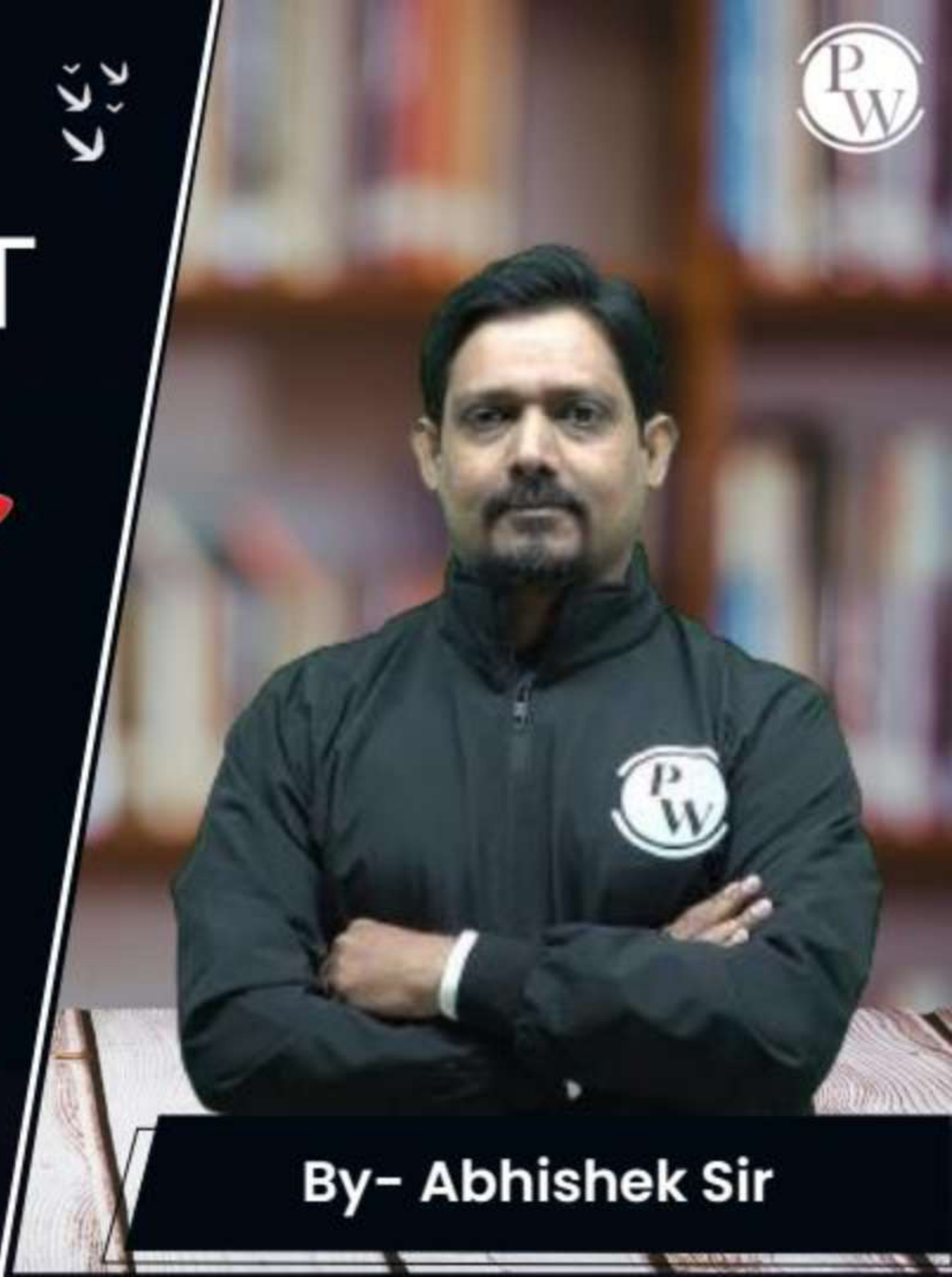
# Computer Science & IT



## C programming

## Control Flow Statement

Lecture No. 04



By- Abhishek Sir

# Recap of Previous Lecture



- Topic — for loop
- Topic — Mathematical
  - $\left\lceil \frac{UB-LB+1}{K} \right\rceil$
  - $\lfloor \log_2 n \rfloor$ - $i=2$
  - $\lfloor \log_2 n \rfloor + 1$ , $i=1$
- Topic — Nested loop
- Topic
- Topic

Slide

# Topics to be Covered



- Topic — while loop
- Topic — do while
- Topic
- Topic
- Topic

Slide



# For loop

- Nested for loop
- While loop
  - Flow chart
  - For loop equivalent to while loop
  - For loop equivalent while loop
  -
-



# While loop

for loop

Initialization

condition

Increment / Decrement

while loop

* entry controlled loop
* we dont know the No. of iteration and depending upon No. of times condition is true the loop will run

Slide



# While loop

```
int i;
for (i=1; i<=n; i++)
    printf("Name");
```

⇒

```
int i;
i=1;
while(i<=n){
    printf("Name");
    i++;
}
```

for equivalent while loop

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# While loop

```
while ( )  {

    printf("Name");

}
```

Condition can't skipped in while

Error



# Do-while loop

* do while is exit control loop

```
do {
    statements;
} while (condition);
```

* condition can't be empty
* Even if condition is false
  the body of loop will execute
  one time
* if condition is false
  for loop - 0 time
  while loop - 0 time
  do while - 1 time

Slide

```
int i;
for (i=1; i<=10; i+=3)
    printf("Name");
```

4

$$\left\lceil \frac{UB - LB + 1}{K} \right\rceil$$

$$= \left\lceil \frac{10-1+1}{3} \right\rceil = \lceil 10/3 \rceil$$

$$= \lceil 3.33 \rceil$$

$$= 4$$

4

```
int i;
i=1;
while (i<=10){
    printf("Name");
    i=i+3;
}
```

1<=10 — ①

4<=10 — ②

7<=10 → ③

10<=10 → ④

4

```
int i;
i=1;
do {
    printf("Name");
    i=i+3;
} while(i<=10);
```

① — i=4 , 4<=10

② — i=7 , 7<=10

③ → i=10 , 10<=10

④ → i=13 , 13<=10 x exit



# Question

Consider the following program segment

```
#include<stdio.h>
int main(){
      int x, i, j;
      x = 0;
      for (i = 0;i<20; i++)
            for(j=i+1;j<18;j++)
                  x++;
      }
```

HW

The value of x after executing the segment

(A) 171 (B) 153 (C) 342 (D) 380

Slide



# While loop

| 6<10 | 7<10 | 8<10 | 9<10 | 10<10 |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | |

```
#include <stdio.h>
int main() {
      int i=2+4%6+9/10;
      while (i<10){
          printf("I am good student");
          i++;
      }
      return 0 ;
      }
```

i = 2+4%6+9/10

= 2+4+0

= 6

The number of times printf statement executed is 4 times

i will be 10

Slide



# Question

Consider the following Program

```
#include<stdio.h>
      int main(){
            int i=1, j=1;
            for(; j; printf("%d%d\t",i,j))
                  j=i++ <= 5;
            return 0 ;
      }
```

(a) 11 22 33 44 5

(b) 21 31 41 51 61

(c) 21 31 41 51 61 70

(d) Infinite loop

HW

break
continue

Slide



# Question

Consider the following C program:

```
#include <stdio.h>
int main(){
      int a = 6;
      int b = 0;
      while(a < 10) {
            a = a / 12 + 1;
            a += b;
      }
      printf("%d", a);
      return 0;
}
```

Which one of the following statements is CORRECT?

(A) The program prints 9 as output

(B) The program prints 10 as output

(C) The program gets stuck in an infinite loop

(D) The program prints 6 as output

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# Question

2024

Consider the following C program:

```
#include <stdio.h>
int main(){
        int a = 6;
        int b = 0;
        while(a < 10) {
                a = a / 12 + 1;
                a += b;
        }
        printf("%d", a);
        return 0;
}
```

a = 6
b = 0
6 < 10 ✓

a = 6/12 + 1
= 0 + 1 = 1

a = a + b = 1 + 0 = 1

Second & third condition same

1 < 10 ✓

a = 1/12 + 1 = 1

a = a + b
= 1 + 0 = 1

1 < 10

Slide

Consider the following C program

```
#include <stdio.h>
int main(){
float sum = 0.0, j = 1.0, i = 2.0;
while (i/j > 0.0625){
      j= j+j;
      sum = sum + i/j;
      printf("%f\n", sum);
}
return 0;
}
```

while loop No. of times execute

The number of times the variable sum will be printed, when the above program is executed, is [5].

lide

Consider the following C program

```
#include <stdio.h>
int main(){
float sum = 0.0, j = 1.0, i = 2.0;
while (i/j > 0.0625){
      j= j+j;
      sum = sum + i/j;
      printf("%f\n", sum);
}
return 0;
}
```

while $(2.0/1.0 > 0.0625)$ — 3

$j = 2.0$

$Sum = 0 + \frac{2.0}{2.0}$

while $(2.0/2.0 > 0.0625)$

$1 > 0.0625$

$j = 4.0$

(2)

while $\left(\frac{2.0}{4.0} > 0.0625\right)$

$0.5 > 0.0625$

$j = 8.0$

(3)

while $\left(\frac{2.0}{8.0} > 0.0625\right)$

$.25 > 0.0625$

$j = 16.0$

(4)

while $\left(\frac{2.0}{16.0} > 0.0625\right)$

$0.125 > 0.0625$

$j = 32.0$

(5)

while $\left(\frac{2.0}{32.0} > 0.0625\right)$

$0.0625 > 0.0625$

false

Consider the following C program

```
#include <stdio.h>
int main(){
float sum = 0.0, j = 1.0, i = 2.0;
while (i/j > 0.0625){
      j= j+j;
      sum = sum + i/j;
      printf("%f\n", sum);
}
return 0;
}
```

The number of times the variable sum will be printed, when the above program is executed, is _____.

Last sum pointed ??

$$Sum = \frac{2.0}{2.0} + \frac{2.0}{4.0} + \frac{2.0}{8.0} + \frac{2.0}{16} + \frac{2.0}{32}$$

$$1 + 0.5 + .25 + .125 + .0625$$

$$\begin{array}{r} 1.0 \\ .5 \\ .25 \\ .125 \\ .0625 \\ \hline 1.9375 \end{array}$$

lide



# Question

Consider the function func shown below:

```
#include <stdio.h>
int main() {
        int num=65;
        int count = 0;
        while (num) {
                count++;
                num>>= 1;
        }
        printf ("%d", count);
}
```

The value printed is 7.

Shift right operator  num/2 (1 shift)

while(65) → count - 1, num = 32

while(32) → 2, num = 16

while(16) → 3, num = 8

while(8) → 4, num = 4

while(4) → 5, num = 2

while(2) → 6, num = 1

while(1) → 7, num = 0

while(0) → false, exit

7

Slide



# 2 mins Summary

Topic — while Loop

Topic — do while

Topic

Topic

Topic

Slide

# THANK - YOU

